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# Active-site geometry, not cofactor chemistry, sets the weak-field magnetosensitivity of neurotransmitter-metabolising flavoenzymes<sup>†</sup>

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That fast recombination and strong exchange coupling suppress weak-field magnetic effects is established in general terms for the radical-pair mechanism (RPM). What a general bound cannot say is where a *particular* enzyme falls — and it is particular enzymes, the flavoenzymes that metabolise monoamine and amino-acid neurotransmitters, that recent proposals of geomagnetic influence on brain chemistry invoke. Using an exact Liouville-space master equation with spin-selective recombination and electronic dephasing, and radical-pair parameters fixed by the crystallographic active-site geometry, we compute the geomagnetic-field ( $50\mu\text{T}$ ) effect on the singlet yield of monoamine oxidase A and B and D-amino-acid oxidase. The response is capped by a purely kinetic ceiling,  $(\omega_B\tau)^2 \approx 8 \times 10^{-7} \%$ , which follows from the picosecond active-site lifetime alone and which no coherent mechanism evades; the computed value is  $1.7 \times 10^{-12} \%$ , verified to be physical rather than numerical by its clean  $B^2$  scaling over two decades of field. Cryptochrome, treated in the identical framework, gives  $+1.0\%$ . The discriminating variable is therefore not the flavin cofactor, which the two share, but the inter-radical separation that the protein scaffold imposes. We further find the lifetime, not the exchange coupling, to be decisive: at  $\tau \approx 10$  ps the response is the same to within a few per cent whether or not the gigahertz exchange coupling is present. Neither a quantum-Zeno route nor the  $\Delta g$  mechanism restores sensitivity. The model also says where to test the null: the active-site singlet–triplet crossing lies at 26.8 mT, so a fine scan of monoamine-oxidase turnover through 10–40 mT, not a coarse sweep, is the discriminating experiment.

The radical-pair mechanism (RPM) is the only experimentally supported route by which a magnetic field as weak as the geomagnetic field ( $\sim 50\mu\text{T}$ ) can alter the outcome of a chemical reaction<sup>1,2</sup>. When a spin-correlated pair of radicals is born in a pure electronic singlet (or triplet) state, the difference in local hyperfine fields drives coherent singlet–triplet interconversion; because recombination is spin-selective, an applied field that re-tunes this interconversion changes the product yields. The mechanism underlies the leading model of avian magnetoreception, in which the photoreceptor cryptochrome hosts a flavin–tryptophan radi-

cal pair<sup>3–5</sup>.

The success of the RPM in cryptochrome has motivated the broader question of whether other flavoproteins, and in particular the enzymes that metabolise neurotransmitters, might be magnetosensitive<sup>6,7</sup>. The question is not purely hypothetical: applied magnetic fields have been reported to alter monoamine-oxidase activity in brain tissue. Rapidly alternating fields potentiate MAO-B with an “active-window” dependence on the field slew rate<sup>8</sup>, and repetitive transcranial magnetic stimulation inhibits MAO-B in synaptosomal preparations<sup>9</sup>; a recent review surveyed the wide range of biological magnetic-field effects attributed to the RPM and noted its applicability to neurochemistry<sup>7</sup>. These reports involve strong or time-varying fields. Whether a *weak, static* field such as the geomagnetic field could bias the catalytic flux of monoamine oxidase (MAO) or D-amino-acid oxidase (DAO) through the radical-pair mechanism—coupling an environmental field directly to monoamine and amino-acid neurotransmitter levels—is the specific, and physiologically consequential, question we address.

This proposal is testable, but a correct test requires three ingre-

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<sup>†</sup> Additional footnotes to the title and authors can be included e.g. ‘Present address:’ or ‘These authors contributed equally to this work’ as above using the symbols: ‡, §, and ¶. Please place the appropriate symbol next to the author’s name and include a \footnotetext entry in the the correct place in the list.

dients that are frequently omitted: a physically realistic radical-pair *lifetime*, the inter-radical *exchange coupling*, and *electronic decoherence*. Weak-field sensitivity is fragile. [The low-field effect arises from the lifting of zero-field degeneracies, which opens additional singlet–triplet pathways<sup>10</sup>, and its magnitude and anisotropy are governed by the interplay of the hyperfine tensors with the applied field<sup>11</sup>; it therefore demands that coherent singlet–triplet mixing, driven by hyperfine couplings of order 1–10 mT, survive long enough and remain undamped by faster competing processes. Treatments that propagate the spin density operator in the coherent limit, neglect the exchange interaction, and assume a microsecond lifetime can report sizeable effects that disappear once realistic kinetics and decoherence are restored.](#)

[Here we](#) That this is so *in general* is not in question, and we do not claim it as a finding. Binhi has recently reviewed the field and, from an analytic treatment of a two-electron–one-nucleus pair, established in general terms both that primary chemical RPM effects in biochemistry are of order  $10^{-3}$ – $10^{-4}$  or smaller, and that the widely assumed radical–electron decoherence times in proteins exceeding 1  $\mu$ s are not substantiated<sup>12</sup>. We take that result as our starting point rather than our conclusion. A general relaxation-limited bound is regime-general by construction: it applies with equal force to cryptochrome, which *is* a functional magnetoreceptor, and to monoamine oxidase, which has been claimed to be one. It cannot by itself adjudicate between them, and it returns no number for any named enzyme. Three things are therefore still missing, and a neurochemical claim needs all three: how far below the general bound a given active site actually falls once its parameters are fixed by its own crystal structure; which physical factor—lifetime, exchange coupling, or decoherence—does the suppressing, since only the decisive factor can be argued about experimentally; and what measurement would overturn the verdict.

Here we supply those three for the neurotransmitter-metabolising flavoenzymes. We compute the magnetic field effect (MFE) on the singlet yield of MAO-A, MAO-B, DAO, and—as a positive [control—cryptochrome](#)control in the identical framework—cryptochrome, using an exact Liouville-space master equation that includes spin-selective recombination and electronic dephasing. [The result is a negative one, stated as a quantitative](#) Three results follow. First, a quantitative enzyme-specific bound: the active-site [radical pairs of the neurotransmitter-metabolising flavoenzymes are insensitive to the geomagnetic field at](#) pairs respond at  $1.7 \times 10^{-12} \%$ , capped robustly by a purely kinetic ceiling  $(\omega_B \tau)^2 \approx 8 \times 10^{-7} \%$  that the picosecond lifetime imposes on its own; cryptochrome, same code and same numerics, returns +1.0%. Second, an attribution: it is the lifetime, not the gigahertz exchange coupling, that is decisive at  $\tau \approx 10$  ps—which relocates the argument from a quantity nobody can measure in this system to one that can be. Third, a falsifiable consequence: the singlet–triplet crossing of the level of  $10^{-12} \%$ ; the picosecond lifetime is the mechanism that robustly enforces this, and the percent-level effects of coherent-limit treatments overestimate the response for this regimeactive-site pair lies at 26.8 mT, so the discriminating experiment is a fine field scan through the tens-of-millitesla range. The contribution of this paper is the bound itself and the physical argument that fixes it is not that weak-field effects are small—that is known—but that magnetosensitivity is decided by the geometry the protein scaffold imposes on the pair rather than by the cofactor it carries, and by how much.

## 1 Methods

### 1.1 Spin Hamiltonian

Each enzyme is modelled as a two-electron radical pair coupled to the dominant hyperfine nuclei. For the flavin-centred radical these are the isoalloxazine ring nitrogens and protons, with isotropic couplings of order 10–20 MHz; the partner radical is identified per enzyme in Table 1. The spin Hamiltonian (in angular-frequency units,  $\hbar = 1$ ) is

$$H = \omega_B (S_{1z} + S_{2z}) + \sum_i \mathbf{S}_1 \cdot \mathbf{A}_i \cdot \mathbf{I}_i + J \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_1 \cdot \mathbf{D} \cdot \mathbf{S}_2, \quad (1)$$

where  $\omega_B = g\mu_B B/\hbar$  is the electron Zeeman frequency at field  $B$  ( $g \approx 2$ ),  $\mathbf{S}_{1,2}$  are the two electron spins,  $\mathbf{A}_i$  are the isotropic hyperfine tensors coupling electron 1 to the dominant ring nuclei  $\mathbf{I}_i$ ,  $J$  is the isotropic inter-radical exchange coupling, and  $\mathbf{D}$  is the electron–electron dipolar tensor. We retain the dominant isotropic hyperfine terms and treat the dipolar coupling in the point-dipole approximation at the inter-radical distance  $r$ . The computational model carries five spins: the two electrons plus three dominant hyperfine-coupled nuclei (two on the flavin radical, one on the partner), giving a 32-dimensional Hilbert space. The conclusions are insensitive to the precise hyperfine assignment, which controls only the field scale at which a *separated* pair would respond, not the verdict for the active-site pairs ([electronic supplementary material, Table S1](#) [ESI](#), [† Table S2](#)).

### 1.2 Master equation

The radical pair is born in the singlet state,  $\rho_0 = P_S/\text{Tr} P_S$ , where  $P_S = \frac{1}{4} - \mathbf{S}_1 \cdot \mathbf{S}_2$  projects onto the electronic singlet and  $P_T = I - P_S$  onto the triplet manifold. Its density operator evolves under

$$\frac{d\rho}{dt} = -i[H, \rho] - \frac{1}{2}\{k_S P_S + k_T P_T, \rho\} + \sum_{i=1,2} \frac{1}{T_2^e} (S_{iz} \rho S_{iz} - \frac{1}{4} \rho). \quad (2)$$

The second term is the Haberkorn spin-selective recombination superoperator<sup>13</sup>, with singlet and triplet recombination rates  $k_S$  and  $k_T$ . The final term is electronic pure dephasing of each electron spin at rate  $1/T_2^e$ , which damps the singlet–triplet coherence that the hyperfine interaction generates. We take  $k_S = k_T = k = 1/\tau$  so that the recombination lifetime  $\tau$  enters as a single rate; making recombination strongly singlet-selective does not enhance the weak-field effect.

The observable is the time-integrated singlet yield,  $\Phi_S = k_S \int_0^\infty \text{Tr}[P_S \rho(t)] dt$ . This steady-state expression is a standard consequence of Eq. (2): writing the Liouvillian  $\mathcal{L}$  for the right-hand side and vectorising the density operator, the integral is performed in closed form,

$$\Phi_S = k_S \text{vec}(P_S)^\dagger (-\mathcal{L})^{-1} \text{vec}(\rho_0), \quad (3)$$

which we evaluate by exact linear solution in the  $32^2$ -dimensional Liouville space; in the coherent, symmetric-recombination limit this reproduces the closed-form Green’s-function singlet yield to machine precision ([electronic supplementary material, S1](#)). The

magnetic field effect is  $\text{MFE}(B) = [\Phi_S(B) - \Phi_S(0)]/\Phi_S(0)$ , reported at  $B = 50\mu\text{T}$ . All four enzymes were propagated with identical numerics; only the per-enzyme parameters differ.

### 1.3 Parameters

The per-enzyme exchange coupling, lifetime, and electronic coherence time are collected in Table 1. The exchange coupling follows the standard exponential distance dependence  $J(r) = J_0 e^{-\beta r}$  with  $\beta \approx 1.4\text{ \AA}^{-1}$ , the standard decay constant for through-protein electron exchange<sup>14,15</sup>; this is an order-of-magnitude estimate, but the conclusions depend only on whether  $|J|$  is large or small compared with the hyperfine scale ( $\sim$ tens of MHz), not on its precise value. At the active-site inter-radical distance of the flavoenzymes ( $r \approx 3.5\text{--}4\text{ \AA}$ ), this gives  $|J| \sim 0.4\text{--}0.75\text{ GHz}$ ; for the spatially separated cryptochrome pair ( $r = 15\text{--}20\text{ \AA}$ ),  $|J| < 1\text{ MHz}$ , effectively zero. That the inter-radical exchange and dipolar couplings quench the weak-field response unless the radicals are separated by more than  $\sim 3.5\text{ nm}$  is an established result of radical-pair compass theory<sup>14</sup>; the present calculation applies it to the active-site geometry of the metabolic enzymes. For the separated cryptochrome pair, spin relaxation analyses place the relevant electronic coherence in the microsecond range<sup>16</sup>, consistent with the coherence-favourable value adopted here.

Lifetimes are fixed by the chemistry. In the flavoenzyme active site the radical pair, if it forms at all, recombines by ultrafast back electron transfer. At the van der Waals donor-acceptor contact of the active site ( $r \approx 3.5\text{ \AA}$ ), empirical electron-transfer rate laws place the transfer rate at  $\sim 10^{11}\text{--}10^{12}\text{ s}^{-1}$ , i.e.  $\tau \sim 1\text{--}10\text{ ps}$ <sup>15</sup>; we adopt the conservative (long) end,  $\tau \sim 10\text{ ps}$  ( $k \sim 10^5\text{ MHz}$ ). The stopped-flow flavin-reduction kinetics measured for MAO<sup>17–19</sup> bound the overall catalytic turnover, not this putative intermediate, and are orders of magnitude slower. The precise value is not critical: Fig. 1a shows the active-site effect stays below  $4 \times 10^{-6}\%$  even for a lifetime  $10^5$  times longer, because the gigahertz exchange coupling—not the lifetime—sets the ceiling. In cryptochrome the electron-transfer chain spatially separates the radicals and the pair survives into the microsecond regime ( $k \sim 1\text{ MHz}$ ,  $\tau \sim 1\mu\text{s}$ )<sup>4,20</sup>. The two regimes also have physically distinct coherence environments. At the active site the radicals share a small, rigid pocket, where the gigahertz exchange and dipolar couplings, a dense local hyperfine bath, and picosecond recombination together hold the electronic coherence to the nanosecond range,  $T_2^e \sim 1\text{ ns}$ . In the spatially separated cryptochrome pair the radicals are mutually isolated and weakly coupled, and spin-relaxation analyses place the electronic coherence in the microsecond range<sup>16</sup>; we adopt a coherence-favourable  $T_2^e$  up to the lifetime as the most optimistic case. These are different physical environments, not a single dephasing mechanism scaled by three orders of magnitude. The disparity is more-over not load-bearing for the null result: the active-site effects are already  $\sim 1.7 \times 10^{-12}\%$  in the coherent limit ( $T_2^e \rightarrow \infty$ , Table 1), so the bound holds for *any* active-site coherence time; for this regime decoherence only tightens it. The electronic-structure inputs—hyperfine magnitudes and inter-radical distances—were taken from semi-empirical GFN2-xTB calculations on the relevant

crystal structures<sup>21</sup> cross-checked against published ENDOR hyperfine couplings for flavin radicals<sup>22</sup>; the MAO active-site geometry is that resolved crystallographically<sup>19</sup>.

## 2 Results

### 2.1 The null result

Table 1 reports the calculated MFE at  $50\mu\text{T}$  for all four enzymes. For MAO-A, MAO-B, and DAO the geomagnetic-field effect on the singlet yield is of order  $10^{-12}\%$  at the physical picosecond lifetime, both in the coherent limit and with realistic electronic decoherence. Even in the most favourable case for an active-site pair—the coherent limit with an unphysically long lifetime—the exchange and dipolar couplings together cap the effect at  $4.0 \times 10^{-6}\%$  (Fig. 1a). We stress that these are model-internal values for the five-spin Hamiltonian of Eq. (1): they quantify the response of the model, not a rigorous bound on the enzyme. **They are, however, physical values and not an artefact of finite-precision arithmetic: the computed MFE follows the perturbative  $B^2$  law to better than 0.3% from  $100\mu\text{T}$  to  $25\text{ mT}$ , and extrapolating that converged slope back to  $50\mu\text{T}$  reproduces the tabulated numbers to within 1.4% (ESI,† Table S1). A numerical floor would exhibit no such scaling.** Because the effect already vanishes in the coherent limit, the active-site null is independent of the assumed electronic coherence time: the value of  $T_2^e$  cannot be the origin of the contrast with cryptochrome, which follows instead from the exchange coupling and lifetime alone. Cryptochrome, by contrast, shows a clear effect of  $+1.0\%$  with decoherence included ( $+2.0\%$  in the coherent limit), of the sign and order of magnitude expected for a functional magnetoreceptor. This is an optimistic-coherence estimate—we assign cryptochrome an electronic coherence time up to its lifetime—so the true value lies below it; measured in-vitro flavoprotein field effects at weak field are typically at or below the one-percent level<sup>2,4</sup>. The robust point is the contrast: the neurotransmitter-metabolising flavoenzymes are magnetically inert at the geomagnetic field by some twelve orders of magnitude relative to cryptochrome.

This bound is consistent with what experiment reports. The one study to scan *static* field strengths on brain monoamine oxidase found no change in either MAO-A or MAO-B activity from  $0.1$  to  $340\text{ mT}$ <sup>8</sup>; the potentiation reported in that work appeared only for rapidly alternating fields, which drive the enzyme through a non-equilibrium, time-dependent mechanism distinct from the weak-static-field RPM analysed here. Our calculation supplies the mechanistic reason for that static-field null and extends it down to the geomagnetic range.

### 2.2 Why the active-site pairs are inert

Figure 1 explains the null result physically. Panel (a) plots the geomagnetic-field  $|\text{MFE}|$  against the radical-pair lifetime  $\tau$  for two geometries: a spatially separated pair with  $J \approx 0$ , and an active-site pair with  $J = 750\text{ MHz}$ . For the separated pair (blue), the effect grows with lifetime and only becomes appreciable once  $\tau$  exceeds roughly  $100\text{ ns}$ , because singlet-triplet interconversion driven by sub-millitesla hyperfine fields needs hundreds of nanoseconds to develop. The flavoenzyme regime ( $\tau \approx 10\text{ ps}$ ,

shaded) sits far to the left, where the effect falls below  $\sim 1.7 \times 10^{-12} \%$  for *both* geometries—at this lifetime the exchange coupling barely matters: picosecond recombination freezes the spin state before any singlet–triplet mixing can occur. Independently, the active-site pair (red), even if given a long lifetime, saturates at  $|MFE| \sim 4.0 \times 10^{-6} \%$  because its large exchange coupling holds the singlet and triplet levels far apart in energy, so a  $50 \mu\text{T}$  field cannot bring them into resonance. Either mechanism—the picosecond lifetime or the gigahertz exchange coupling—suffices to suppress the effect; the flavoenzymes have both.

Panel (b) shows that for the separated geometry electronic decoherence tightens the bound. For the most favourable separated-pair case ( $\tau = 1 \mu\text{s}$ ,  $J \approx 0$ ),  $|MFE|$  collapses as the electronic coherence time  $T_2^e$  falls below  $\sim 100$  ns; recovering a weak-field effect requires *both*  $\tau \gtrsim 100$  ns and  $T_2^e \gtrsim 100$  ns. The active-site value  $T_2^e \sim 1$  ns lies two orders of magnitude below this threshold.

### 2.3 Why coherent-limit treatments overestimate the effect

If the dephasing term of Eq. (2) is dropped *and* a microsecond lifetime is assigned to the active-site pair, the same model returns percent-level MFEs for the flavoenzymes. Both assumptions are required: the long lifetime lets singlet–triplet mixing run to completion, and omitting decoherence prevents that mixing from dephasing. Neither holds for an active-site radical pair, where back electron transfer is picosecond-fast and hyperfine-induced dephasing is nanosecond-fast. A coherent-limit treatment that makes these two assumptions for an active-site pair will therefore overestimate the weak-field effect by orders of magnitude. We do not claim that any particular earlier study is in error—a given treatment may have addressed a different geometry, mechanism, or parameter regime—but rather state the methodological boundary: an RPM estimate for an active-site cofactor radical must propagate the physical lifetime and electronic decoherence, or it will overstate the effect by orders of magnitude.

### 2.4 Mechanistic caveat

A further point strengthens, rather than weakens, the negative conclusion. It is not established that MAO catalysis proceeds through a discrete radical-pair intermediate at all. Multireference CASSCF(2,2)/NEVPT2 calculations on a lumiflavin–methylamine model<sup>23–25</sup> give only weak diradical character along the hydrogen-transfer coordinate (Fig. 2): the diradical index is  $y_0 = 0.03$ – $0.09$  and the singlet–triplet gap is  $3.2$ – $4.1$  eV across the relevant  $d(\text{N5} \cdots \text{H})$  range. Such values favour a polar or concerted hydride/proton-coupled pathway over genuine radical-pair formation<sup>18</sup>, in agreement with QM/MM studies of MAO catalysis that find no radical intermediate and support a concerted asynchronous polar nucleophilic mechanism<sup>26</sup>. This calculation is deliberately suggestive rather than definitive: it samples three geometries with a minimal CAS(2,2) active space on a reduced lumiflavin–methylamine model, and does not by itself exclude a radical-pair intermediate in the full enzyme. We present it only as an independent, consistent indication; the central result of the paper—the weak-field bound—does not depend on it, since that bound holds *whether or not* an active-site radical

pair forms. If one does not, the RPM does not apply; if one does, its geomagnetic effect is the negligible value computed above.

## 3 Discussion

The picture that emerges is that magnetosensitivity is not a generic property of flavoprotein cofactor radicals; it must be engineered. Cryptochrome achieves it with a purpose-built electron-transfer chain—the tryptophan triad—that separates the two radicals by  $15$ – $20$  Å, quenching the exchange coupling and extending the lifetime into the microsecond regime where a  $50 \mu\text{T}$  field can act<sup>3–5,20</sup>. The neurotransmitter-metabolising flavoenzymes have the opposite geometry: their chemistry happens in a tight active site, where any radical pair recombines in picoseconds and feels a gigahertz exchange splitting. Each factor alone forbids a weak-field effect; together they place it near  $10^{-12} \%$ , with realistic electronic decoherence pushing it lower still. The geomagnetic field cannot meaningfully bias the singlet yield of an active-site radical pair in MAO-A, MAO-B, or DAO, and proposals that it modulates monoamine or amino-acid neurotransmitter levels through the RPM are not supported.

### 3.1 Relation to the general relaxation-limited bound

That fast recombination and strong exchange coupling suppress weak-field effects is established, and Binhi’s recent review gives it in closed form for a two-electron–one-nucleus pair, together with a catalogue of protein relaxation channels that places generic biological RPM effects at  $10^{-3}$ – $10^{-4}$  or below and rejects the routine assumption of microsecond electron decoherence in proteins<sup>12</sup>. Our results are consistent with that analysis, and where they overlap they corroborate it from an independent direction: the nanosecond  $T_2^e$  assigned here to a shared, rigid active-site pocket is reached from the active-site geometry rather than from a relaxation-mechanism catalogue. Three things nevertheless separate the present calculation from the general result, and a claim about neurochemistry needs all three.

First, a regime-general bound cannot discriminate. Whatever forbids weak-field sensitivity in monoamine oxidase must be reconciled with cryptochrome, where weak-field sensitivity is observed<sup>4</sup>. The present calculation returns  $1.7 \times 10^{-12} \%$  and  $+1.0 \%$  for the two *in the same code with the same numerics and the same five-spin model*; what separates them is the inter-radical separation the protein scaffold imposes, not the flavin cofactor they share. That contrast, rather than the smallness of either number, is the result of this paper.

Second, these enzymes fall far below the general estimate, by an amount only enzyme-specific parameters can supply. The generic  $10^{-3}$ – $10^{-4}$  figure is a *relaxation-limited* bound. What actually binds at a flavoenzyme active site is *kinetic*:  $(\omega_B \tau)^2 \approx 7.8 \times 10^{-7} \%$ , fixed by the picosecond lifetime that the crystallographic van der Waals donor–acceptor contact enforces through the Moser–Dutton rate law<sup>15</sup>. This ceiling is some four orders of magnitude tighter than the generic figure, is independent of every relaxation parameter, and the computed response lies a further five orders below it. Neither number is available from a general analysis, because neither survives without the crystal structure.



Third, a general bound identifies no experiment. Because the suppression here is lifetime-dominated rather than exchange-dominated—a distinction the general treatment does not draw—the falsifying observation is specific and is stated below: a resonant feature at the 26.8 mT singlet–triplet crossing would show that the active-site pair lives orders of magnitude longer than assumed.

Two limitations of scope should be stated plainly. First, the computed observable is the time-integrated singlet yield of an isolated pair, not a turnover number. We take  $k_S = k_T$ , for which the total recombination flux is field-independent by construction; the bound is therefore on the spin-selective branching that any RPM-based effect on catalysis would have to exploit, not on the enzymatic rate itself. Since  $\Phi_S$  is the quantity any downstream spin-selective step must read, bounding it does bound the achievable RPM effect—but connecting the two quantitatively would require a kinetic scheme with distinct singlet and triplet product channels, which we have not constructed. Second, we have not simulated radical-triad or scavenger mechanisms, in which a third radical restores magnetosensitivity to a pair that is insensitive on its own<sup>27</sup>. Such schemes require the pair to survive long enough to encounter the third spin, which is implausible on a 10 ps timescale, but we have not tested this and it remains the most plausible route by which our conclusion could fail for a longer-lived intermediate.

One recent result deserves explicit attention. It has been shown that a tightly bound radical pair—one with a large exchange coupling, of the kind found at an enzyme active site—can nonetheless become sensitive to an Earth-strength field when its recombination is strongly spin-asymmetric, through a quantum Zeno mechanism<sup>28</sup>. This appears to contradict the present conclusion, so we tested it directly: for the monoamine-oxidase parameters ( $|J| = 750$  MHz,  $\tau \approx 10$  ps) we recomputed the MFE with strongly asymmetric recombination, varying  $k_S/k_T$  over six orders of magnitude. The geomagnetic-field MFE remained below  $10^{-7}\%$  for every asymmetry, and even at a 200-fold stronger field (10 mT) it did not exceed  $1.5 \times 10^{-3}\%$  (Supporting Information, Table S2 ESI,† Table S3). The quantum Zeno mechanism cannot rescue magnetosensitivity here because it requires the recombination to act over a window in which spin evolution can be projected; the picosecond active-site lifetime is too short for any such window to open. The Zeno route to weak-field sensitivity in tightly bound pairs therefore applies to longer-lived species (for example a flavin–superoxide pair<sup>28,29</sup>) but not to the picosecond active-site radical pair of a metabolic flavoenzyme.

The null is likewise insensitive to the form of the electronic decoherence. For a tightly bound pair the dominant channel is expected to be singlet–triplet dephasing driven by fluctuations in the exchange coupling, whose Lindblad operator ( $\propto P_S - P_T$ ) differs from the single-spin channel used above. Adding this channel at any rate up to and beyond the exchange coupling leaves the geomagnetic-field MFE far below the level of interest (Supporting Information, Table S3 ESI,† Table S4). We emphasise that this is an empirical result for the two channels examined, not a general theorem: because the MFE is a *ratio*, a dissipative channel acts on  $\Phi_S$  at zero and at finite field alike and can in principle *increase* it. Our own model provides an example—for a long-lived ( $\tau = 1 \mu\text{s}$ ) pair

at  $J = 750$  MHz, introducing  $T_2^e = 1$  ns changes the geomagnetic-field MFE from  $-1.4 \times 10^{-6}\%$  to  $+3.6 \times 10^{-4}\%$ , a change of sign and two orders of magnitude. The picosecond active-site regime studied here is not subject to this, but the choice of dephasing model must be justified rather than assumed to be conservative.

We also tested the  $\Delta g$  mechanism, the one coherent channel that survives when the exchange coupling greatly exceeds the hyperfine scale and therefore the natural candidate for rescuing a strongly coupled pair. At the realistic value  $\Delta g \simeq 10^{-3}$  the geomagnetic-field MFE is unchanged, and even at  $\Delta g = 3 \times 10^{-2}$  it reaches only  $3 \times 10^{-10}\%$  (Supporting Information ESI,† Table S5). The underlying reason is a scale hierarchy that no coherent mechanism evades: the recombination rate  $1/\tau = 10^5$  MHz exceeds the largest internal coupling ( $|D| = 1214$  MHz) by more than two orders of magnitude, and the field enters only through  $\omega_B$ , so the response is bounded by  $(\omega_B \tau)^2 \approx 2 \times 10^{-10}$  ( $\omega_B \tau$ )<sup>2</sup>  $\approx 7.8 \times 10^{-9}$ —that is,  $2 \times 10^{-8}\%$   $7.8 \times 10^{-7}\%$ —however the remaining couplings are arranged, *some four*. This ceiling lies *some five* orders of magnitude above the computed response and still *eight orders below any measurable level about five orders below the  $\sim 0.1\%$  level at which in-vitro magnetic field effects are detectable*. This is why the active-site and separated geometries agree to within a few per cent at  $\tau = 10$  ps despite differing by three orders of magnitude in  $J$ .

This bound concerns the RPM specifically. It does not exclude all conceivable magnetic influences on neurochemistry—for example field effects mediated by reactive-oxygen-species partitioning<sup>29,30</sup> or other non-radical-pair channels—but those would require their own mechanism and evidence and are not addressed by a singlet-yield calculation. A separate and unrelated proposal, Fisher’s suggestion that Posner molecules store quantum information in <sup>31</sup>P nuclear spins<sup>31</sup>, invokes nuclear rather than electronic spin coherence; it is a different mechanism, equally unconfirmed, and our electronic-spin bound says nothing for or against it.

The conclusion is experimentally falsifiable, and the model identifies where to look. The electron Zeeman splitting matches the active-site exchange coupling ( $J = 750$  MHz) at  $B = 26.8$  mT, so the singlet–triplet level crossing of the active-site pair lies not at Tesla fields but inside the tens-of-millitesla range. In our model a *long-lived* pair with this exchange coupling shows a large resonant response there (tens of per cent at  $\tau = 1 \mu\text{s}$ ), while the physical picosecond pair shows  $\sim 5 \times 10^{-7}\%$ . The discriminating experiment is therefore not a coarse 0–50 mT sweep but a fine scan through the crossing: measuring steady-state MAO turnover from 10 to 40 mT in  $\sim 1$  mT steps should return a null result. A resonant feature near 27 mT would not require physics beyond the standard RPM—it would indicate that the active-site radical pair lives orders of magnitude longer than assumed here, which is the single assumption on which our conclusion rests.

### 3.2 Relation to a previous preprint by the authors

An earlier version of this work was deposited by the authors on Research Square and has not been peer reviewed<sup>32</sup>. The present manuscript differs from it in substance, not only in presentation. (i) *The observable has changed*. The preprint reported a bound on “catalytic flux”, a quantity a singlet-yield calculation does not

deliver; here the computed observable is the time-integrated singlet yield and the limits of that scope are stated explicitly. (ii) *The mechanistic attribution has changed*, which is why the title has changed. The preprint attributed the suppression to electronic decoherence and exchange coupling; the present calculation identifies the picosecond lifetime as decisive—at  $\tau \approx 10$  ps the response is unchanged to within a few per cent when the gigahertz exchange coupling is removed, and the null already holds in the coherent limit, so decoherence cannot be its origin. (iii) *The bound is now quantitative and verified*. The preprint quoted  $< 10^{-4}\%$ ; here we give  $1.7 \times 10^{-12}\%$ , the analytic ceiling  $(\omega_B \tau)^2$ , and a  $B^2$ -scaling test establishing that the value is physical rather than numerical (ESI† Table S1). (iv) *Four robustness analyses have been added* that the preprint did not contain: the quantum Zeno mechanism<sup>28</sup>, singlet–triplet dephasing, the  $\Delta g$  mechanism, and a multireference CASSCF(2,2)/NEVPT2 test of whether a radical pair forms at all. (v) *An over-strong claim has been withdrawn*. The preprint asserted that percent-level results in the prior literature are “computational artifacts”; that assertion is replaced here by a statement of the methodological requirement, since we cannot audit parameter choices made for a different geometry or mechanism.

## 4 Conclusions

We have computed an exact That fast recombination and strong exchange coupling suppress weak-field magnetosensitivity is known in general terms<sup>12</sup>. What has been missing is where named enzymes fall, which factor puts them there, and what would show otherwise. Using an exact Liouville-space master-equation bound, including electronic decoherence, on the master equation with electronic decoherence and radical-pair parameters fixed by crystallographic active-site geometry, we find the geomagnetic-field sensitivity of three neurotransmitter-metabolising flavoenzymes. The bound is a negative result: the weak-field effect on the singlet yield of their putative effect on the singlet yield of the putative active-site radical pair is of order  $10^{-12}\%$ , enforced independently by a gigahertz active-site exchange coupling and by picosecond recombination, and reduced further by nanosecond electronic decoherence. Only the spatially separated radical pair of cryptochrome supports weak-field sensitivity. Percent-level effects from coherent-limit treatments that omit decoherence and assume microsecond lifetimes overestimate the response for an active-site pair by radical pairs of MAO-A, MAO-B and DAO to be  $1.7 \times 10^{-12}\%$ , capped robustly at  $(\omega_B \tau)^2 \approx 7.8 \times 10^{-7}\%$  by the picosecond lifetime alone—four orders of magnitude. The physical bounds reported here delimit, quantitatively, where the radical-pair mechanism can and cannot operate in brain biochemistry. below the generic relaxation-limited estimate for a biological radical pair, and reached without reference to any relaxation parameter. Cryptochrome, in the same model, returns +1.0%. The decisive variable is therefore the inter-radical separation the protein scaffold imposes and not the flavin cofactor the two enzymes share: magnetosensitivity in a flavoprotein is a property of geometry, which must be built, rather than of cofactor chemistry, which comes for free. Within the active site the suppression is dominated by the lifetime rather than by the exchange coupling, which relocates the argument onto a measurable quantity: the singlet–triplet crossing at 26.8 mT is where a longer-lived pair would announce itself, and a fine field scan there is the experiment that would overturn this conclusion.

## Author contributions

Conceptualization, H.W.; methodology, H.W.; software, H.W.; validation, H.W. and T.T.; formal analysis, H.W.; writing—original draft preparation, H.W. All authors have read and agreed to the published version of the manuscript.

## Conflicts of interest

There are no conflicts to declare.

## Data availability

The source code, parameter files, and analysis scripts supporting this article are openly available in the qbscreen Python package at [https://github.com/deeptell-inc/brain\\_protein\\_screening](https://github.com/deeptell-inc/brain_protein_screening) under the MIT licence. This includes the Liouville-space master-equation solver, the per-enzyme magnetic-field-effect driver, and an automated test suite. The suite pins the values of Table 1 to within 2% by direct recomputation, and separately checks the solver against a closed-form coherent-limit expression; we, and verifies the  $B^2$  scaling of the computed MFE reported in the ESI† (Table S1). We note that the latter closed-form check shares the eigenbasis convention of the solver and therefore validates the Liouville-space inversion rather than the unit conventions; the  $B^2$  test does not, and is in that sense the independent one. A versioned snapshot will be archived on Zenodo with a citable DOI upon acceptance.

## Acknowledgements

## Notes and references

For the reference section, the style file `rsc.bst` can be used to generate the correct reference style.<sup>§</sup>

1. Citations should appear here in the format A. Name, B. Name and C. Name, *Journal Title*, 2000, **35**, 3523;
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Table 1 Per-enzyme radical-pair parameters and calculated MFE at the geomagnetic field ( $B = 50 \mu\text{T}$ ).  $r$  is the inter-radical distance,  $J$  the exchange coupling,  $\tau$  the recombination lifetime, and  $T_2^e$  the electronic coherence time. “Coherent” omits the dephasing term of Eq. (2); “realistic” includes it.

| Enzyme | Partner radical                | $r$     | $J$     | $D$       | $\tau$           | $T_2^e$ | MFE <sub>coh</sub>      | MFE <sub>real</sub>     |
|--------|--------------------------------|---------|---------|-----------|------------------|---------|-------------------------|-------------------------|
| MAO-A  | serotonin amine radical cation | 3.5 Å   | 750 MHz | −1214 MHz | ~10 ps           | 1 ns    | $1.7 \times 10^{-12}\%$ | $1.6 \times 10^{-12}\%$ |
| MAO-B  | dopamine amine radical cation  | 3.5 Å   | 750 MHz | −1214 MHz | ~10 ps           | 1 ns    | $1.7 \times 10^{-12}\%$ | $1.6 \times 10^{-12}\%$ |
| DAO    | D-amino-acid radical           | ~4 Å    | 400 MHz | −813 MHz  | ~10 ps           | 1 ns    | $5.6 \times 10^{-13}\%$ | $5.0 \times 10^{-13}\%$ |
| CRY    | Trp radical cation (Trp triad) | 15–20 Å | < 1 MHz | −9.7 MHz  | ~1 $\mu\text{s}$ | 1000 ns | +2.0%                   | +1.0%                   |

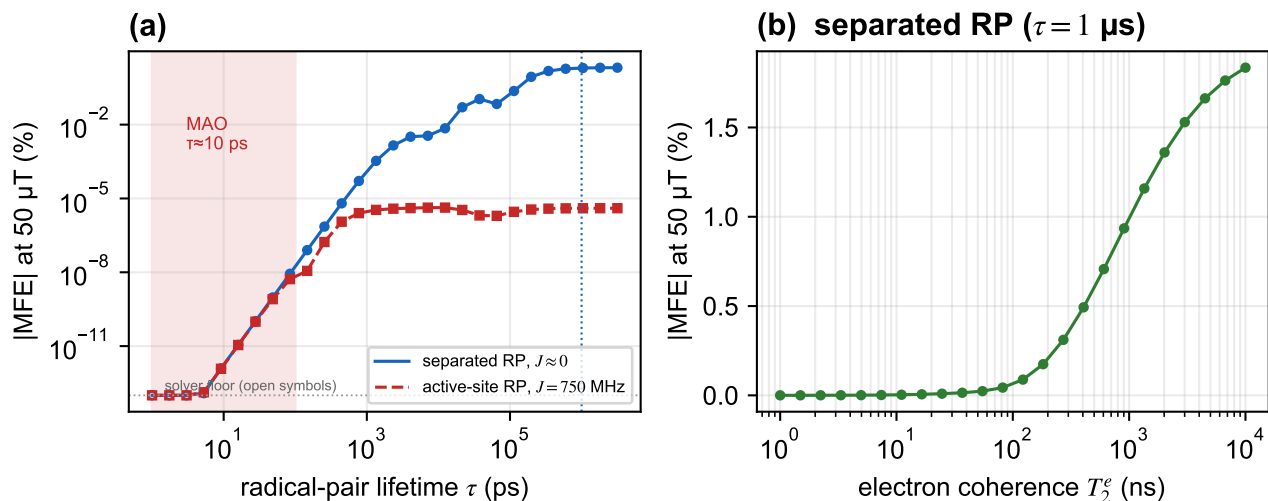


Fig. 1 Calculated geomagnetic-field sensitivity as a function of radical-pair kinetics and electronic coherence. All curves are computed from the Liouville-space master equation of Eq. (2) at  $B = 50 \mu\text{T}$ . (a) Magnitude of the MFE versus radical-pair lifetime  $\tau$  for a spatially separated pair ( $J \approx 0$ ,  $D = -9.7$  MHz, blue circles) and an active-site pair ( $J = 750$  MHz,  $D = -1214$  MHz, red squares), both in the coherent limit ( $T_2^e \rightarrow \infty$ ). Each geometry carries its own point-dipole coupling;  $J$  and  $D$  are both functions of  $r$  and are not varied independently. The separated pair requires  $\tau \gtrsim 40$  ns for an appreciable effect; the active-site pair saturates at  $|\text{MFE}| \sim 4.0 \times 10^{-6}\%$  at long lifetime. The shaded band marks the flavoenzyme regime ( $\tau \approx 10$  ps), where both geometries give  $\sim 1.7 \times 10^{-12}\%$ —the lifetime, not the exchange coupling, is what suppresses the response. Points below the solver floor (dotted line) are drawn as open symbols; the same clamp is applied to both series. (b) Magnitude of the MFE versus electronic coherence time  $T_2^e$  for the optimal separated pair ( $\tau = 1 \mu\text{s}$ ,  $J \approx 0$ ): a weak-field effect survives only when  $T_2^e \gtrsim 180$  ns. The active-site value  $T_2^e \sim 1$  ns lies far below this threshold.

CASSCF(2,2)/NEVPT2: Diradical character along H-transfer coordinate

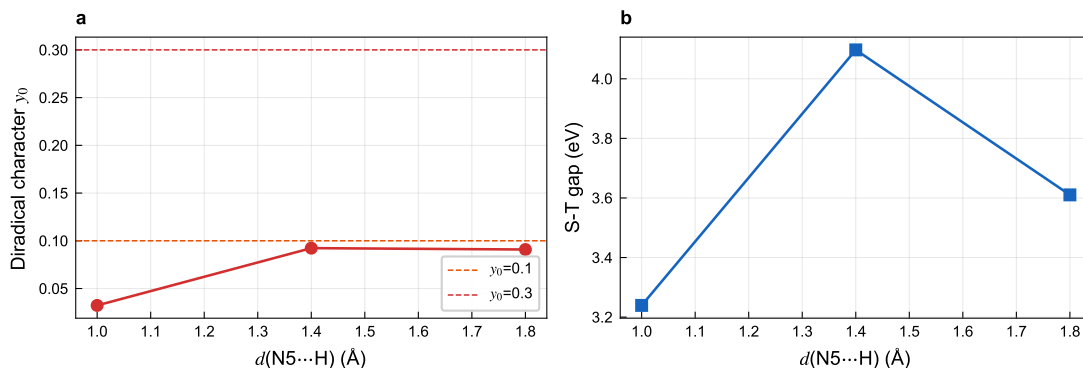


Fig. 2 Calculated diradical character of the MAO transition state. CASSCF(2,2)/NEVPT2 results on a lumiflavin–methylamine model along the hydrogen-transfer coordinate  $d(\text{N5} \cdots \text{H})$ . (a) Diradical character index  $y_0$  stays at 0.03–0.09, well below the  $y_0 = 0.3$  guide line, indicating weak open-shell character. (b) The singlet–triplet gap remains 3.2–4.1 eV throughout. Both quantities are inconsistent with formation of a low-lying, spin-correlated radical pair, favouring a polar/concerted mechanism.

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